FCC-μe Higgs production

Leading-order cross sections and positive-muon polarization

2 November 2026

Scope. Full Standard Model tree-level matrix elements for four final states:

$$\begin{aligned} \text{CC}: &\quad e^-\mu^+ \to \nu_e \bar{\nu}_\mu H, \quad e^-\mu^+ \to \nu_e \bar{\nu}_\mu HH, \\ \text{NC}: &\quad e^-\mu^+ \to e^-\mu^+ H, \quad e^-\mu^+ \to e^-\mu^+ HH. \end{aligned}$$

No imposed generation kinematic cuts; unpolarized e^- beam; $P_{\mu^+} = 0, 20, 40, 60, 80, 100\%$ with $P_{\mu^+} = 100\%$ right-helicity antimuons; $\sqrt{s} = 1$ and 2 TeV, with $(E_e, E_{\mu^+}) = (125, 2000)$ and $(200, 5000)$ GeV, respectively

$\sqrt{s}$	Channel	$\sigma(0)$ [fb]	$\sigma(80\%)$ [fb]	$\sigma(100\%)$ [fb]
1 TeV	CC H	210.540 ± 0.209	378.570 ± 0.243	421.150 ± 0.433
1 TeV	CC HH	0.074384 ± 0.000127	0.133230 ± 0.000230	0.148239 ± 0.000236
1 TeV	NC H	20.835 ± 0.025	24.476 ± 0.015	25.402 ± 0.018
1 TeV	NC HH	0.009620 ± 0.000015	0.011308 ± 0.000017	0.011723 ± 0.000018
2 TeV	CC H	386.010 ± 0.303	695.000 ± 0.614	772.690 ± 0.517
2 TeV	CC HH	0.418510 ± 0.000657	0.754495 ± 0.001277	0.839521 ± 0.001397
2 TeV	NC H	38.991 ± 0.030	45.829 ± 0.055	47.642 ± 0.033
2 TeV	NC HH	0.054712 ± 0.000085	0.064299 ± 0.000096	0.066770 ± 0.000103

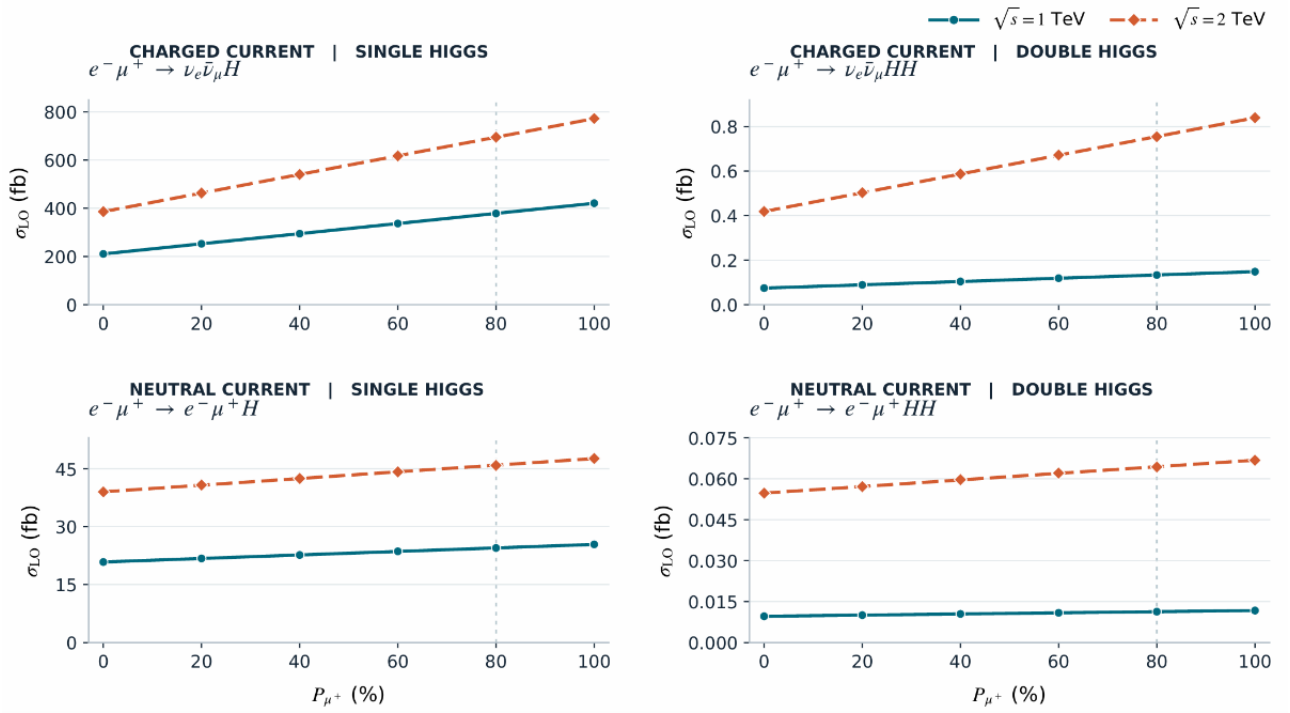


Figure 1: Leading-order Standard Model cross sections for single- and double-Higgs-boson production in μ^+e^- collisions as functions of the longitudinal polarization of the positive-muon beam, P_{μ^+}. The four processes considered are $\mu^+e^- \to \bar{\nu}_\mu \nu_e H$, $\mu^+e^- \to \bar{\nu}_\mu \nu_e HH$ (charged current), and $\mu^+e^- \to \mu^+e^- H$, $\mu^+e^- \to \mu^+e^- HH$ (neutral current). Results are shown for center-of-mass energies of $\sqrt{s} = 1$ and 2 TeV, with an unpolarized electron beam and P_{μ^+} varying from 0% to 100%. The calculations employ the full tree-level Standard Model matrix elements, without imposing kinematic phase-space cuts. Cross sections are expressed in femtobarns (fb). Error bars represent the Monte Carlo integration uncertainties.

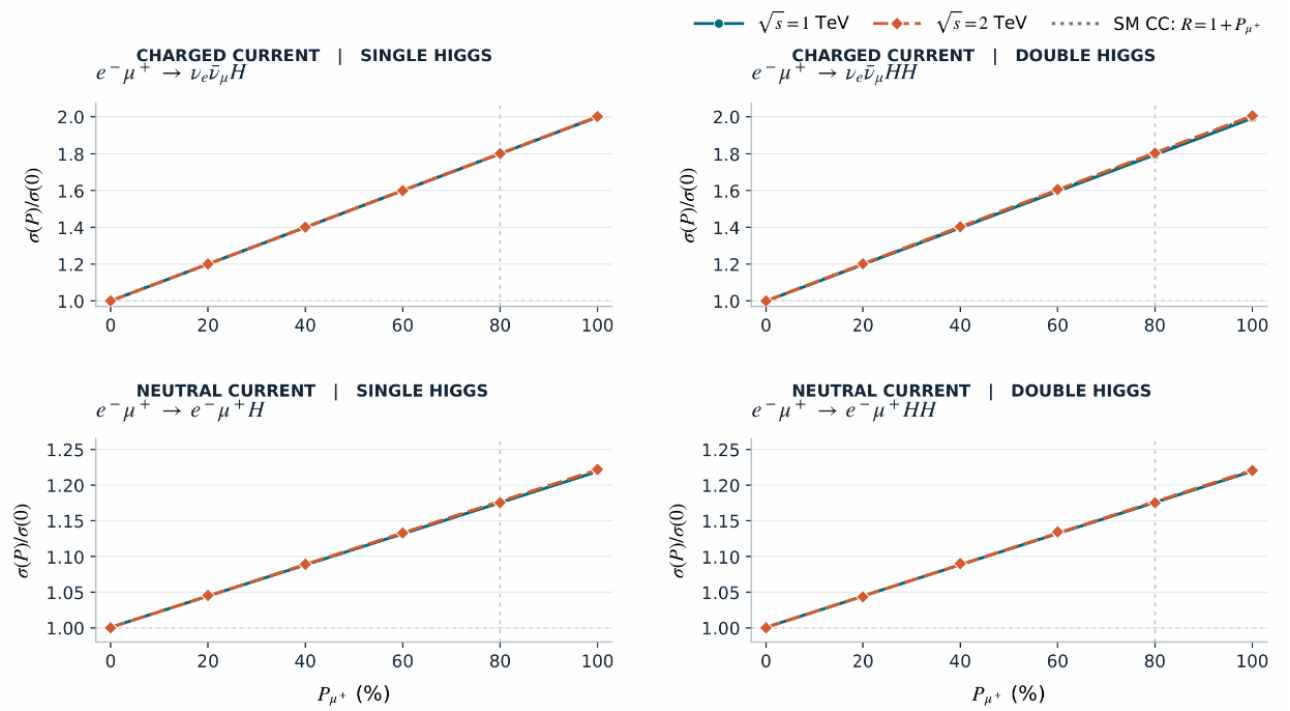


Figure 2: Dependence of the normalized Higgs-production cross sections on the longitudinal positive-muon beam polarization in μ^+e^- collisions. The polarization enhancement factor is defined as $R(P_{\mu^+}) = \sigma(P_{\mu^+})/\sigma(0)$, where $\sigma(0)$ denotes the corresponding unpolarized cross section. Results are presented for charged-current and neutral-current single- and double-Higgs production at $\sqrt{s} = 1$ and 2 TeV.

Inclusive cross sections

$\sqrt{s}$ [TeV]	$P_{\mu+}$ [%]	σ(CC H) [fb]	σ(CC HH) [fb]	σ(NC H) [fb]	σ(NC HH) [fb]
1 TeV	0	210.540 ± 0.209	0.074384 ± 0.000127	20.835 ± 0.025	0.009620 ± 0.000015
1 TeV	20	252.710 ± 0.158	0.088979 ± 0.000149	21.770 ± 0.015	0.010045 ± 0.000017
1 TeV	40	294.730 ± 0.271	0.104016 ± 0.000189	22.660 ± 0.014	0.010487 ± 0.000016
1 TeV	60	336.830 ± 0.199	0.118825 ± 0.000205	23.611 ± 0.019	0.010895 ± 0.000015
1 TeV	80	378.570 ± 0.243	0.133230 ± 0.000230	24.476 ± 0.015	0.011308 ± 0.000017
1 TeV	100	421.150 ± 0.433	0.148239 ± 0.000236	25.402 ± 0.018	0.011723 ± 0.000018
2 TeV	0	386.010 ± 0.303	0.418510 ± 0.000657	38.991 ± 0.030	0.054712 ± 0.000085
2 TeV	20	463.490 ± 0.399	0.502952 ± 0.000846	40.760 ± 0.027	0.057089 ± 0.000087
2 TeV	40	540.620 ± 0.371	0.587089 ± 0.001004	42.460 ± 0.045	0.059636 ± 0.000086
2 TeV	60	617.270 ± 0.428	0.671917 ± 0.001152	44.171 ± 0.042	0.062066 ± 0.000087
2 TeV	80	695.000 ± 0.614	0.754495 ± 0.001277	45.829 ± 0.055	0.064299 ± 0.000096
2 TeV	100	772.690 ± 0.517	0.839521 ± 0.001397	47.642 ± 0.033	0.066770 ± 0.000103

Polarization slope and linearity

We fit the absolute cross section at each energy to

$$\sigma(P_{\mu^+}) = a + b\,p, \qquad p \equiv P_{\mu^+}/100, \qquad A \equiv b/a,$$

using weights $1/\delta\sigma_i^2$. Quoted A errors use the fitted covariance matrix and reported integration errors without systematics.

$\sqrt{s}$ [TeV]	Channel	$A = b/a$	χ^2/ndof	p-value
1	CC H	0.9982 ± 0.0020	1.77/4	0.779
1	CC HH	0.9934 ± 0.0040	2.84/4	0.585
1	NC H	0.2181 ± 0.0012	5.08/4	0.279
1	NC HH	0.2186 ± 0.0022	1.93/4	0.749
2	CC H	1.0007 ± 0.0018	2.38/4	0.667
2	CC HH	1.0053 ± 0.0039	0.94/4	0.919
2	NC H	0.2208 ± 0.0011	3.40/4	0.493
2	NC HH	0.2204 ± 0.0022	3.36/4	0.500